Building a Controlled SOC Lab: Network Design, Isolation & Safe Lab Management

Building a Segmented SOC Lab: Designing a Safe Environment for Detection, Investigation and Attack Simulation

`Designing a realistic security operations environment while keeping attack simulations isolated, observable, and manageable.

- requirements
- threat model
- network architecture
- virtualization
- segmentation
- isolation
- snapshots
- resource constraints
- design decisions
- The difference is design / architecture
- what to architecture ? Network, Snapshot automation

> Setting up a couple of VMs isn't the same as building a lab environment

> The lab was designed around constrained compute and storage resources, so services were consolidated where practical while security boundaries were maintained through network segmentation.

Requirements :
- iso (kali, win 10, win server, metasploitble2) [links]
- wazuh agent, suricata, GNS3, FortiOS7.6 image, ... [links]
- procmon, sysmon, ... [links]

Deciding Network design

I have thought that there are many scenarios where I would want the environment to be separate isolated networks improving security, and other times I might need VMs to communicate for lab purpose, so this leaves us with a network design ambiguity/uncertainty . so

what I wanted to do is to have in some scenarios bridges connected to the same network and in other isolated, also I should be able to do this easily .

There are multiple tools and methods to get this done, but some more efficient in this project then others, I could have used ip command program but it's temporary and doesn't give me the switching behavior I want everything would have to be manually done (get's the work done but not really useful here), the other option is using nmcli directly but also one time fix network no different scenario handling option which leave us to the automation method using yaml config files to set the network architectures desired, instead of bridges preforming the routing and I have to reboot each time change topology.

Using the valuable AI we cam up with an Idea which is creating network profiles in which all desired connections' scenarios are implemented and make a custom script that handles these network profiles and give them NetworkManager eventually creating the desired connections.

```
Your lab configuration
       │
       │ YAML
       ▼
lab-manager script
       │
       │ nmcli commands
       ▼
NetworkManager
       │
       │ connection profiles
       ▼
Linux bridges / interfaces
       │
       ▼
libvirt / VM virtual NICs
       │
       ▼
VM
```

Testing bridges

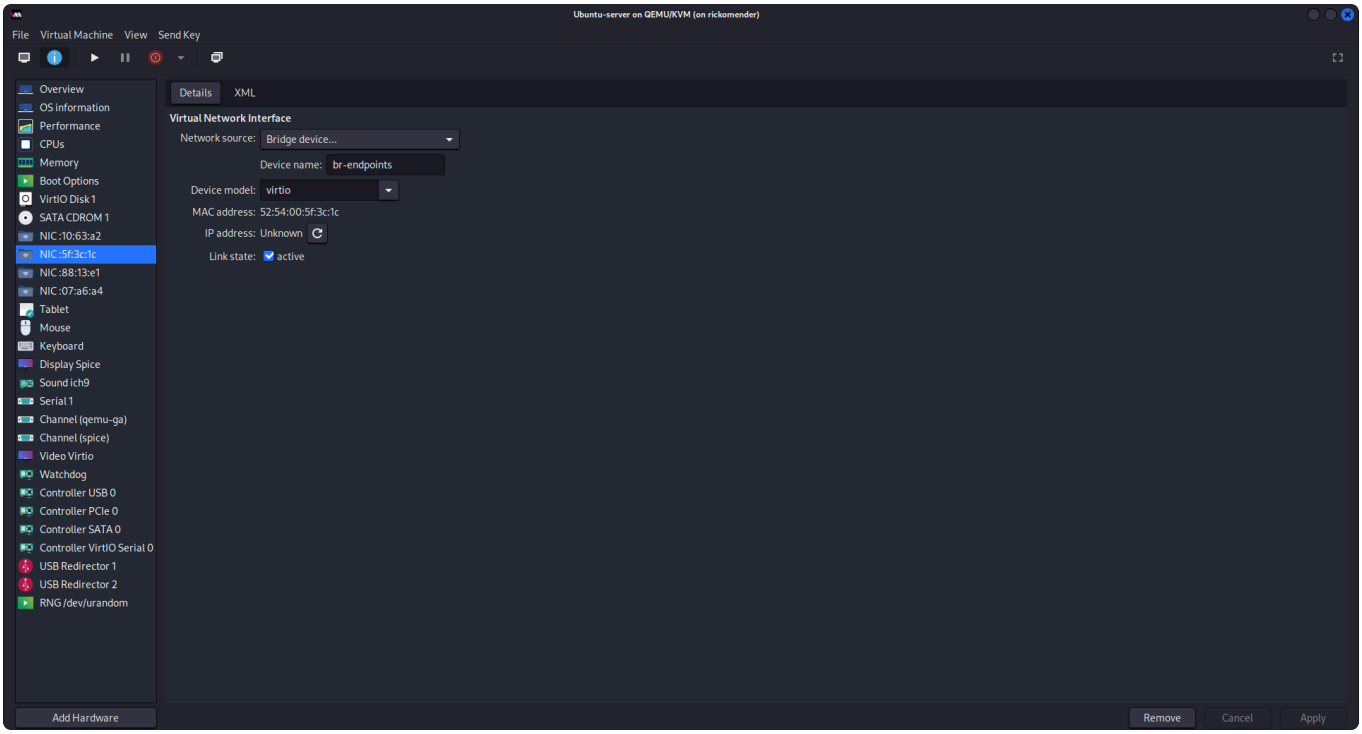
first you will have to test the bridges manually and make sure that the network is working as intended, (try all different networks' topologies)

```
ip link add <br-name> type bridge
```

repeat with desired number of bridges

create multiple NIC hardware for a

then edit the actual VMs using the installed hypervisor, in my case I use QEMU/KVM
so I config the VM from virt manager either from GUI



or through the terminal :

```
virsh edit Ubuntu-server
```

then inside the VM xml config file, search for `<interface type='>` change
`